JAIN DEEMED-TO-BE UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND IT

DEPARTMENT OF CS & IT
Programme: MCA

Activity-1
Cloud Analytics

25MCASC302

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Semester: 3rd Semester

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DESIGN AND DEPLOYMENT OF A PERSONAL PORTFOLIO WEB APPLICATION

1. Introduction

This report documents the design, development and deployment of a personal professional portfolio website. The portfolio presents the student's profile, education, technical skills, projects, experience and contact details in one accessible web application. The project was created to meet the Activity 1 requirement of designing a portfolio, maintaining the complete project in GitHub and publishing a working live link. The solution is intentionally simple, responsive and easy for recruiters or academic evaluators to navigate.

Project objectives

- Create a clear online portfolio that communicates professional background and technical strengths.
- Organize the project source code in an accessible GitHub repository.
- Use a modern frontend workflow for a maintainable and efficient application.
- Deploy the application through an automated GitHub Actions workflow to Microsoft Azure App Service.
- Provide a live portfolio link for final submission in the designated Google Sheet.

Portfolio structure

The visible navigation and page content are arranged around Home, About, Education, Skills, Projects, Experience and Contact. This structure lets a visitor move from an introduction to evidence of skills and work, then to contact channels.

2. Design and Development

The interface uses a single-page portfolio layout with section-based navigation. The design gives priority to readability: clear headings, concise content blocks, cards for skills and projects, and consistent spacing. It is designed for both desktop and smaller screens.

Design approach

- User-focused information order: introduction first, followed by background, skills, projects and contact options.
- Consistent visual hierarchy through headings, cards and grouped sections.
- Professional content presentation with direct links to GitHub, LinkedIn and email.
- A contact form provides a simple way for visitors to send a message.

Development approach

The portfolio was built as a React application using Vite. React supports a component-based approach, allowing the page sections and interface elements to be structured cleanly. Vite provides a fast development server and production build process. The deployment configuration uses Node.js 22 LTS to build and host the application on Azure App Service.

3. Technologies, Tools and Platforms

The portfolio uses a compact modern web-development stack. Each tool was selected for a clear role: React supports the interactive interface, Vite supports development and production builds, GitHub stores the project, and Azure provides the live hosting environment.

Area	Technology
Frontend	React, HTML, CSS and JavaScript
Backend	Node.js with Express.js
Build Tool	Vite
Server runtime	Node.js 22 LTS
Version control	Git and GitHub
Automation	GitHub Actions build and deploy workflow
Cloud Platform	Microsoft Azure App Service

4. Development and Implementation Approach

The portfolio was developed as a frontend React application. It uses reusable components for the navigation bar, hero, about, skills, projects, experience, contact area and footer. This component-based structure makes the interface easier to maintain and lets individual sections be updated without rewriting the full page.

Skills, experience and project details are stored separately from the presentation components. The application uses a lightweight custom router for project case-study pages, CSS variables for consistent visual styling, and reduced-motion support so animations respect user accessibility preferences.

Vite is used to run the local development environment and create the production build. The built portfolio is maintained in GitHub and deployed to Azure App Service through the configured GitHub Actions workflow, giving a clear path from source changes to the live website.

5. Key Features and Functionalities

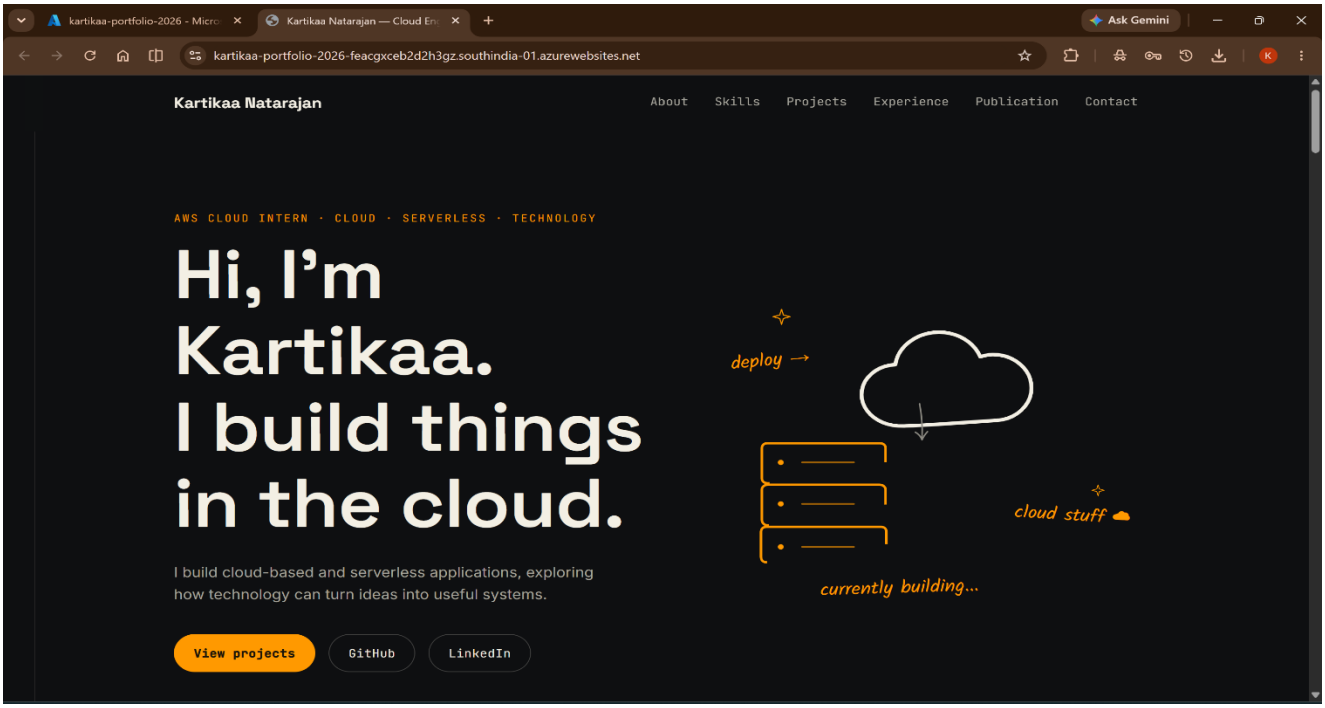
- Home section with an introduction and portfolio navigation.
- About and education information to provide professional context.
- Skills section showing AWS, Python, SQL, SAP ABAP, Git/GitHub and Linux.
- Project cards and dedicated case-study pages presenting selected work.
- Experience section showing AWS Cloud Intern and Associate Analyst roles.
- Contact section with email, LinkedIn, GitHub links and a contact form.
- Automated build and deployment through GitHub Actions to Azure App Service.

Challenges faced and solutions

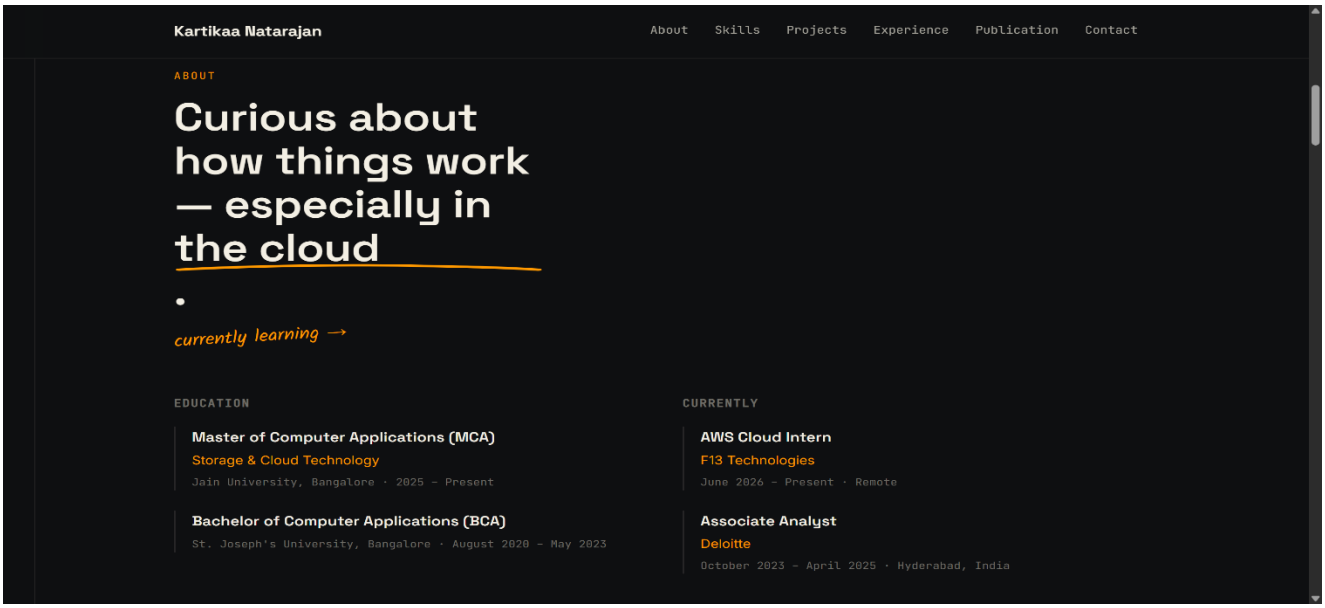
- **Build permission issue**: GitHub Actions initially reported “vite: Permission denied”. The build command was adjusted to run Vite through Node, avoiding reliance on the Linux executable permission.
- **Deployment configuration**: the Azure deployment required the correct App Service workflow, Node 22 LTS environment and publish-profile configuration. The successful GitHub Actions run confirms the build and deploy stages completed.
- **Maintainability:** reusable components, separate data files and a lightweight router keep the growing portfolio organised and easier to update.

6. Portfolio Screenshots: Overview

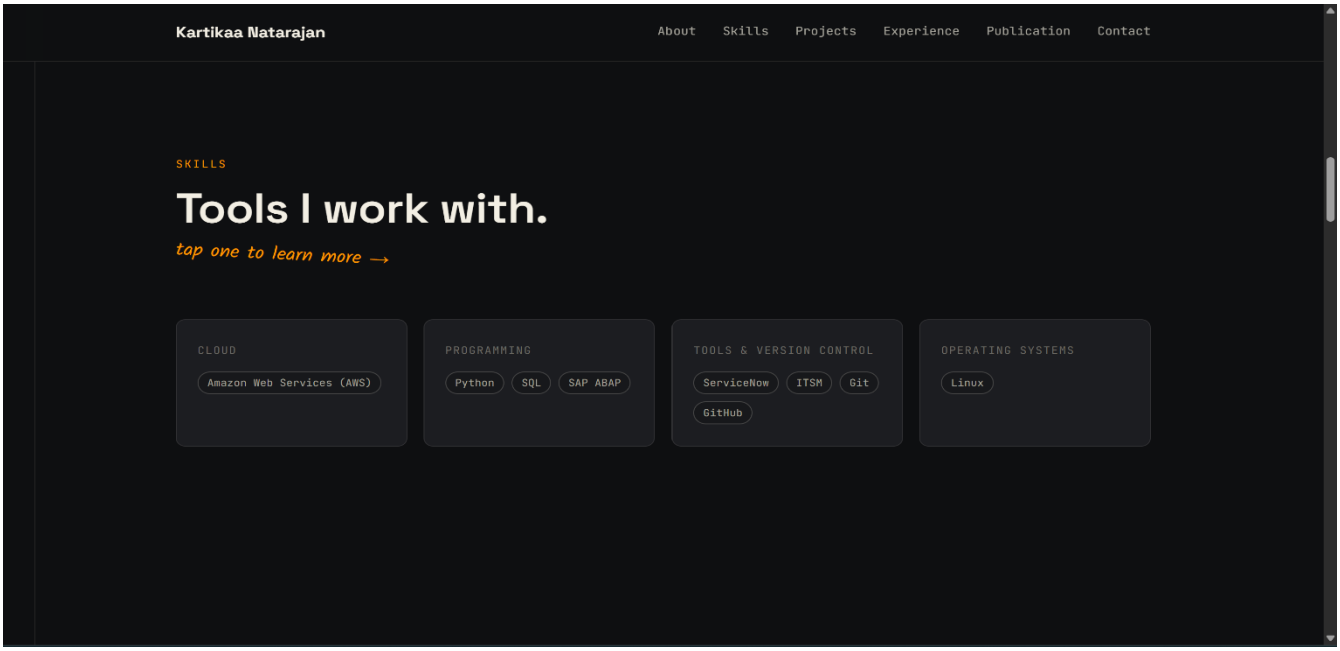
Portfolio home / landing page with introduction and navigation



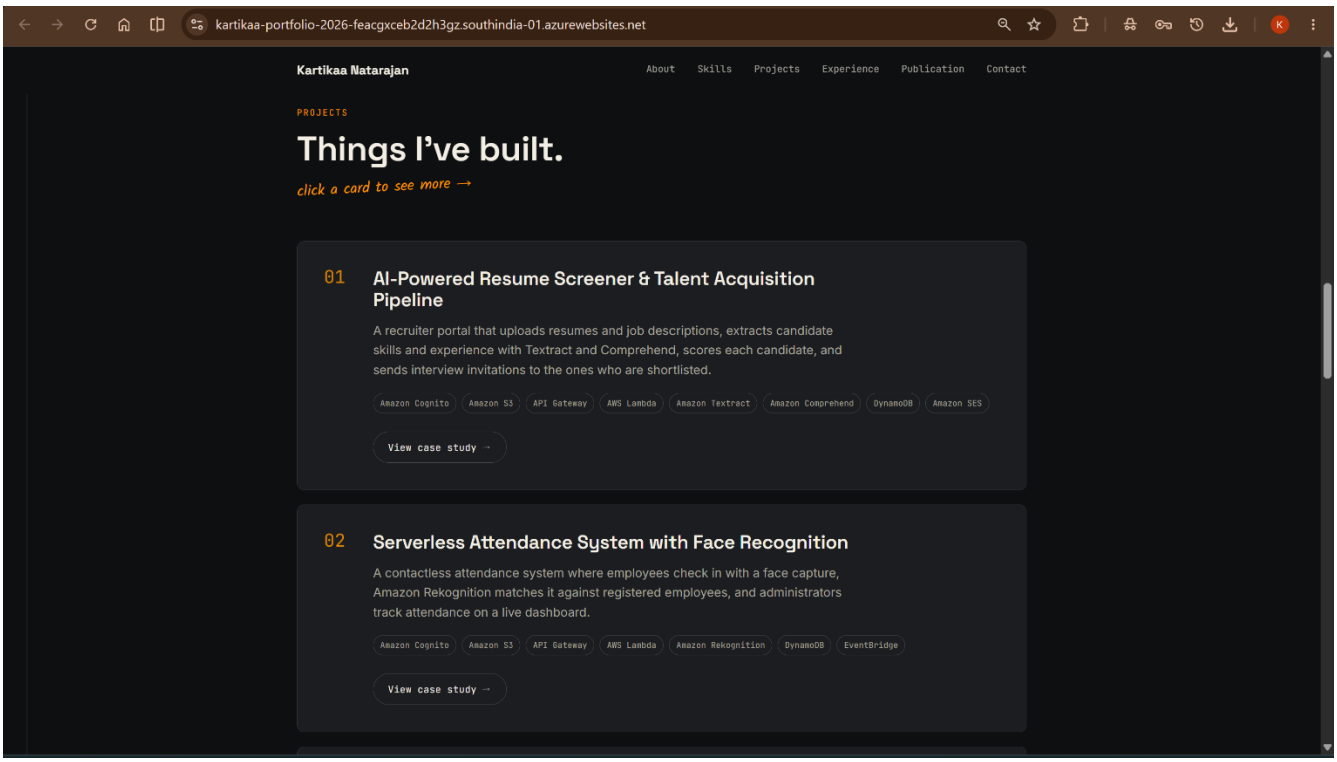
About and education section



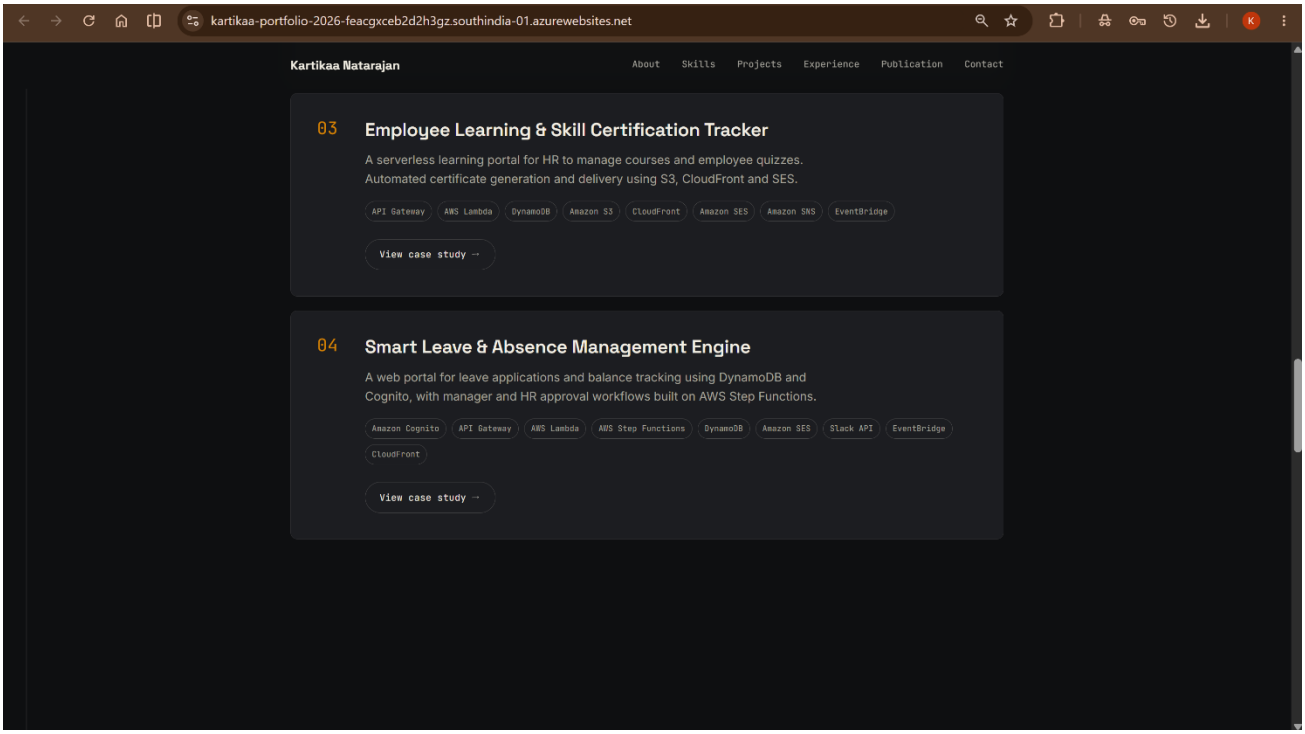
Skills section showing technical competencies



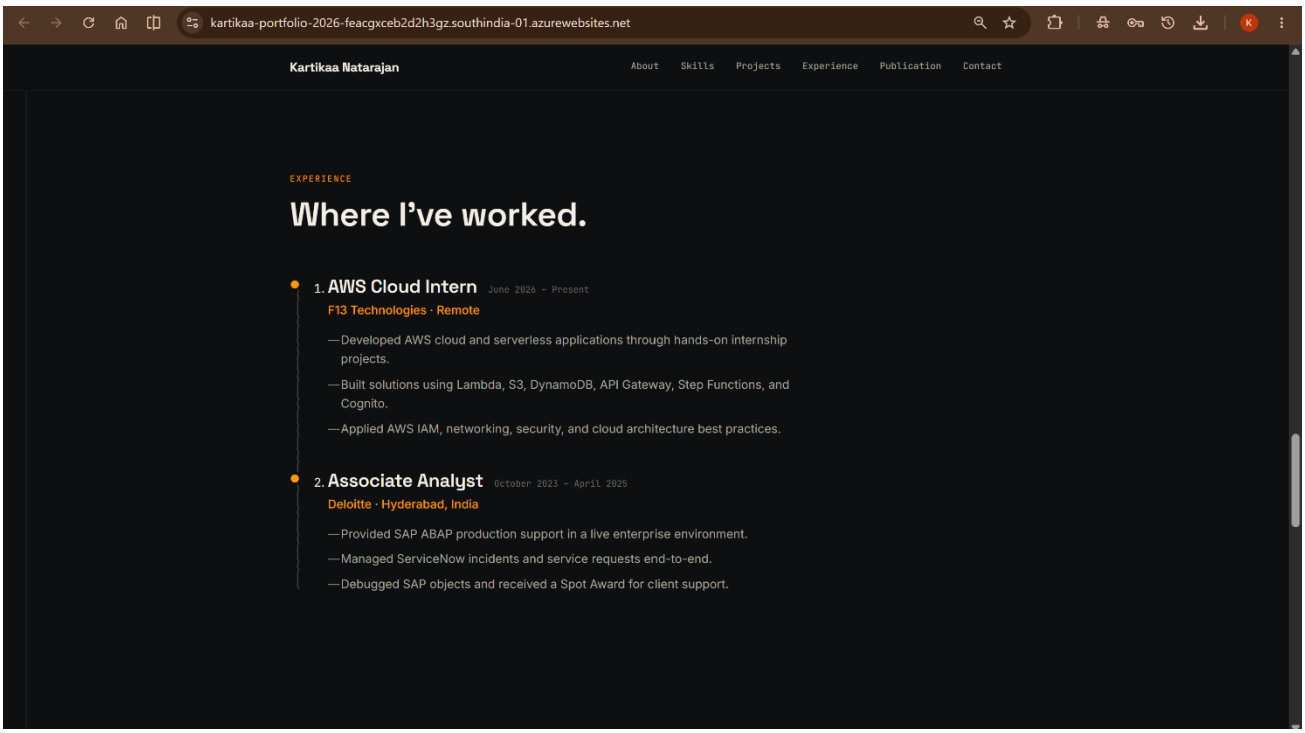
Projects section - first set of project cards



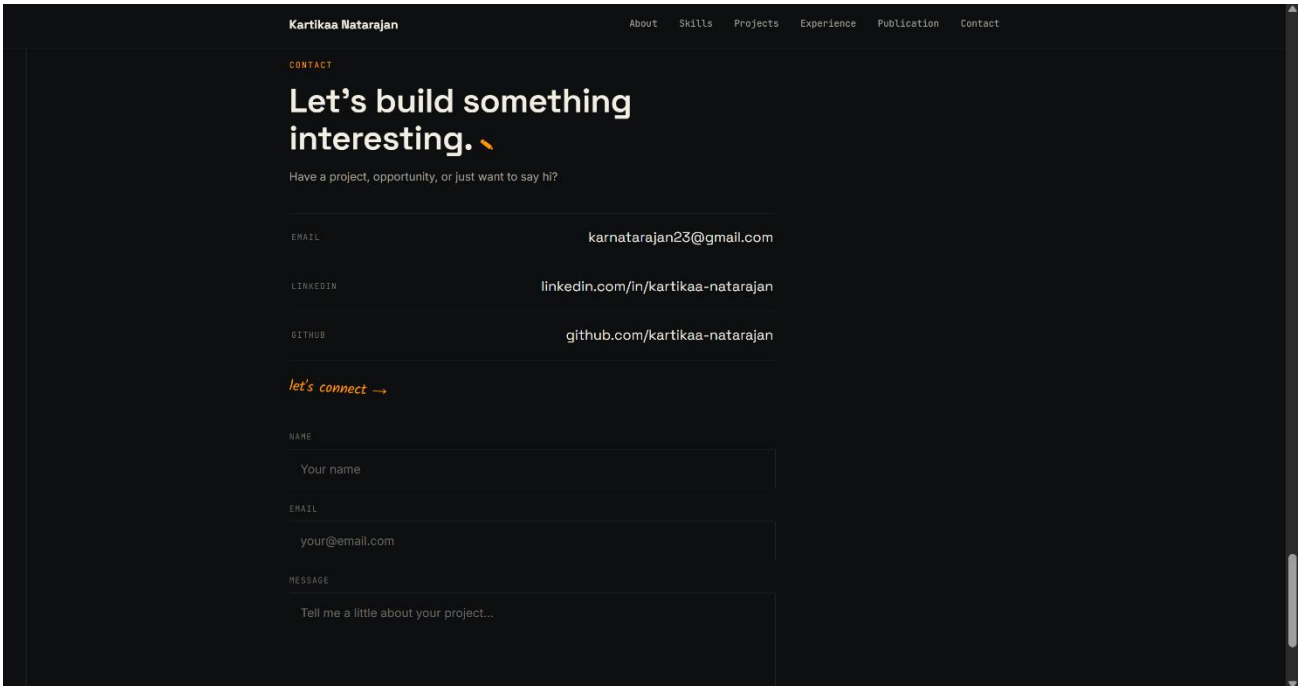
Projects section - additional project cards



Experience section



Contact section with direct links and contact form

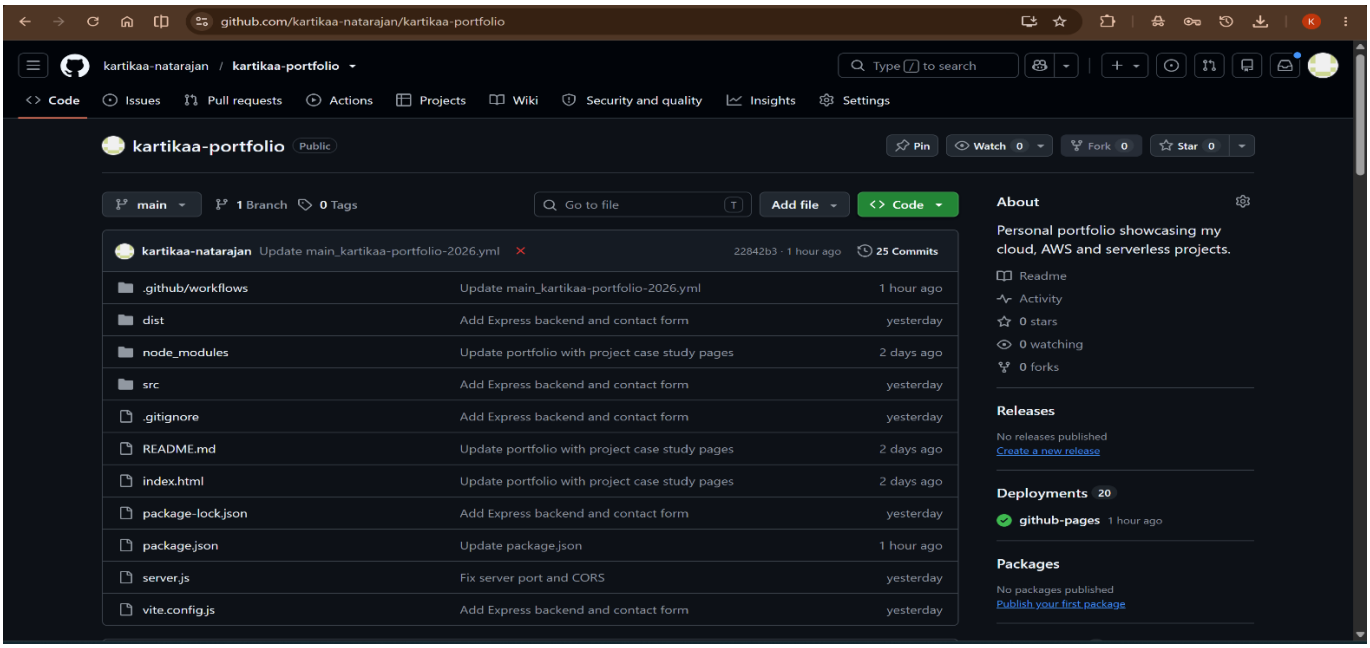


8. GitHub Repository and CI/CD Evidence

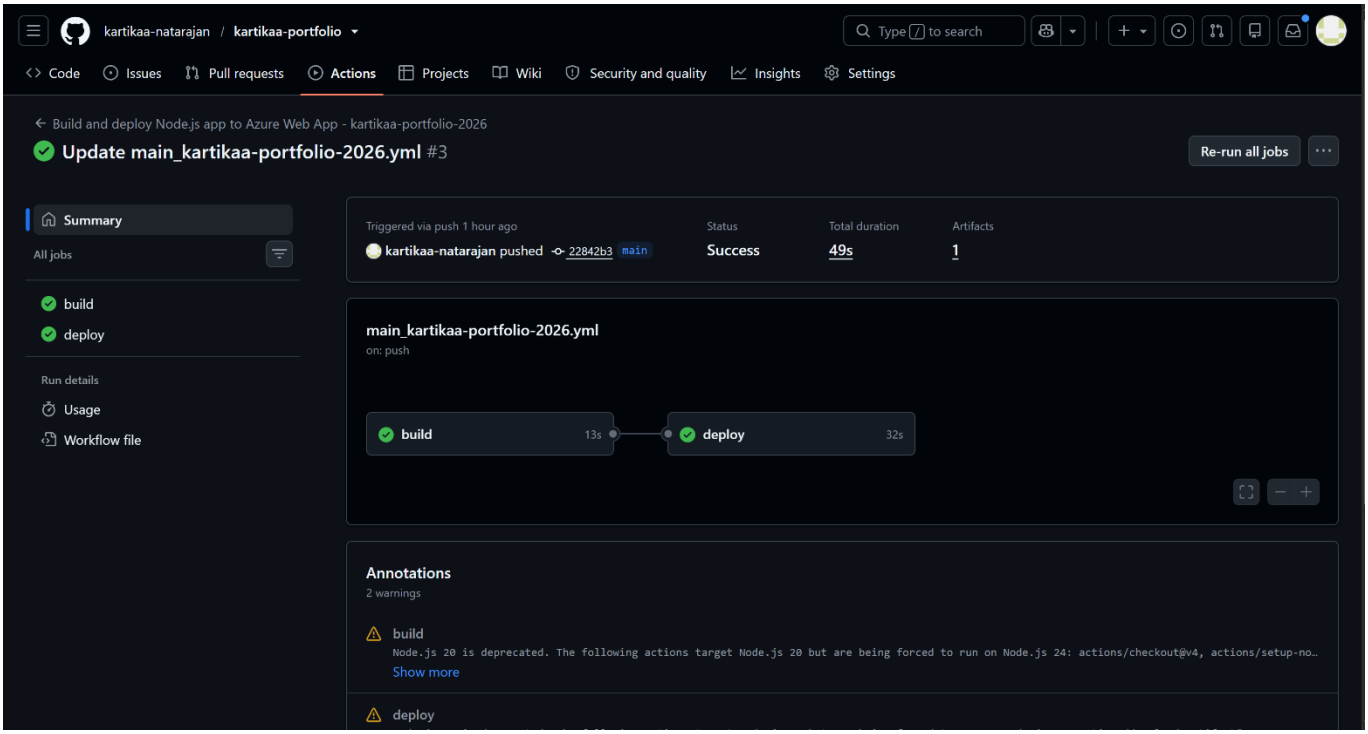
The complete portfolio project is maintained in the GitHub repository **kartikaa-natarajan/kartikaa-portfolio**. The repository view shows the project structure, including the source directory, distribution/build content, package configuration, server file and workflow-related files. Keeping these resources together supports organized maintenance and future updates.

The workflow uses GitHub Actions to build the React/Vite project and deploy it to Azure App Service. The configured deployment target is **kartikaa-portfolio-2026**, with Node.js 22 LTS used in the workflow environment.

GitHub repository structure for the portfolio project



Successful GitHub Actions build and deployment workflow



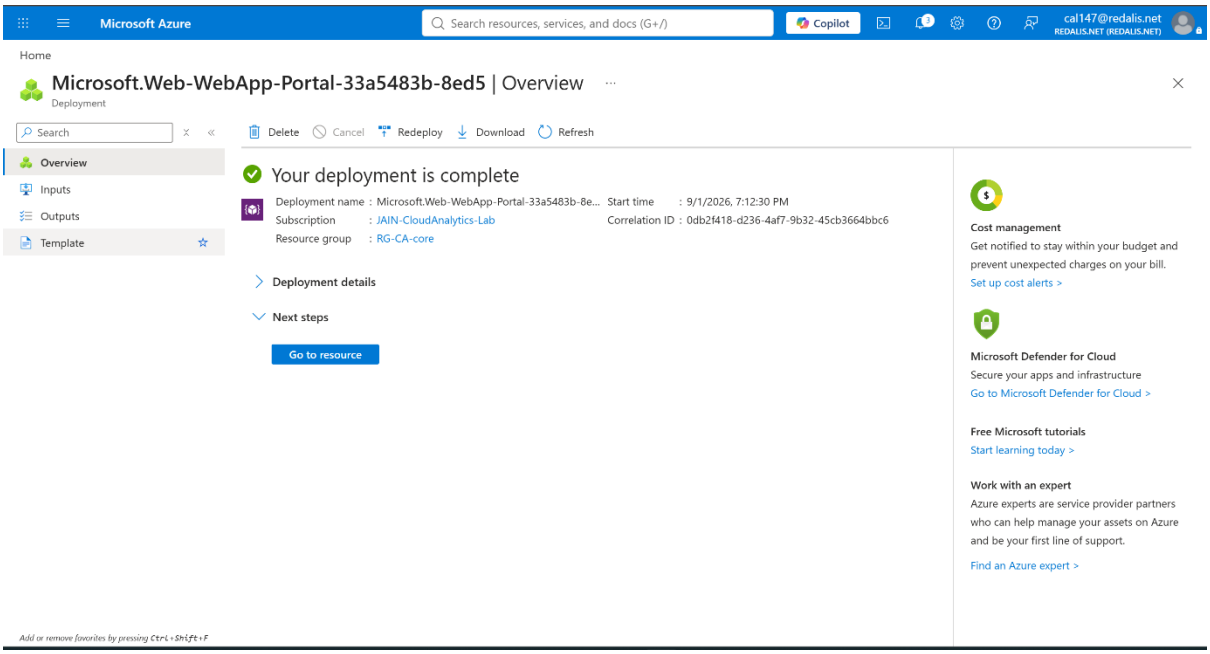
9. Azure Deployment

Azure App Service deployment

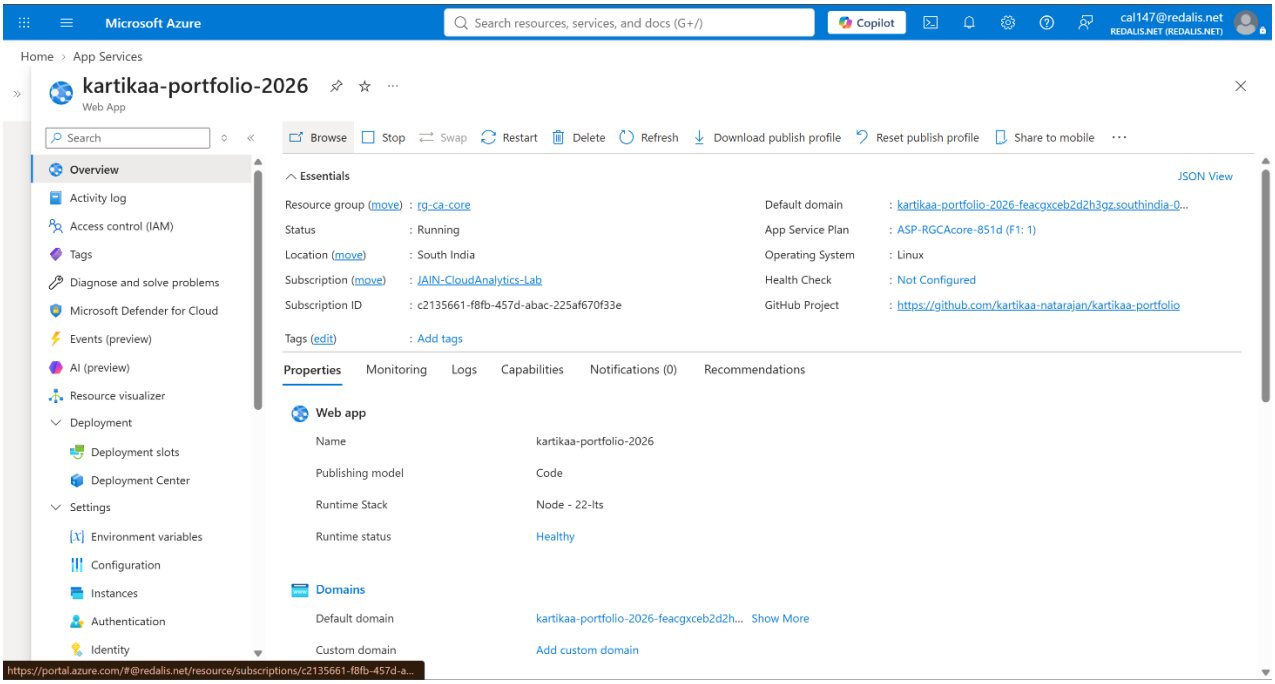
The portfolio is deployed to Microsoft Azure App Service under the application name kartikaa-portfolio-2026. Azure App Service provides the hosted environment for the live portfolio, while GitHub Actions provides the automated CI/CD path from repository updates to deployment.

This deployment approach separates development from hosting: source changes are managed in GitHub, the workflow runs the build process, and Azure serves the completed application. The successful workflow evidence confirms the configured pipeline completed its build and deploy stages.

Successful deployment of the resource



Azure App Service overview showing kartikaa-portfolio-2026 running on Linux with Node 22 LTS.



10. Conclusion

The development of this portfolio successfully fulfilled the objectives of Activity 1 by creating a professional, responsive and accessible personal portfolio that presents academic background, technical skills, projects, professional experience and contact information in a structured manner. The project provided an opportunity to apply modern web development concepts while creating a practical platform for showcasing professional capabilities.

The portfolio was developed using **React and Vite**, following a component-based approach that makes the application organized, maintainable and easier to extend. Features such as section-based navigation, project case studies, skills presentation, experience details, external profile links and a contact form were implemented to provide a complete user experience.

Git and GitHub were used for version control and project management, ensuring that the complete source code and supporting configuration files are maintained in an accessible repository. **GitHub Actions** was integrated to automate the build and deployment process, providing a continuous workflow whenever changes are pushed to the repository. The application was successfully deployed using **Microsoft Azure App Service with Node.js 22 LTS**, resulting in a working live portfolio.

During development, a build permission issue was encountered in the GitHub Actions environment, where Vite initially returned a “Permission denied” error. This was resolved by modifying the build command to invoke Vite through Node, allowing the project to build successfully. The experience provided practical exposure to troubleshooting CI/CD workflows and understanding the relationship between application code, build tools and cloud deployment environments.

11. Submission Links

The completed portfolio project is maintained in the GitHub repository, which contains the source code, project configuration, build files and deployment workflow.

GitHub Repository:
https://github.com/kartikaa-natarajan/kartikaa-portfolio

The portfolio has been successfully deployed and is accessible through Microsoft Azure App Service.

Live Portfolio:
https://kartikaa-portfolio-2026-feacgxceb2d2h3gz.southindia-01.azurewebsites.net

Submission:
Both the GitHub repository link and the live portfolio link should be entered in the designated Google Sheet as part of the Activity 1 submission. The live link has been verified to open the deployed portfolio successfully.